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### Systems and methods for rotating products during processing

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#### Abstract

An exemplary embodiment of the present disclosure provides a system for rotating a product during processing comprising a shackle system comprising a trolley, a rotator, and a rotating shackle and a track wherein the trolley is coupled to the track and configured to undergo translational motion along the track causing the shackle system to move along the track and wherein the rotator is configured to undergo rotational kinematics causing rotation of the rotating shackle.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/086,652 filed on Oct. 2, 2020 and U.S. Provisional Application Ser. No. 63/212,185 filed on Jun. 18, 2021, which are incorporated herein by reference in their entirety as if fully set forth below.

### FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to processing systems and methods and more particularly to systems and methods for rotating food products during processing.

### BACKGROUND

(2) Reducing or precluding cross-contamination of products during processing is desirable. For example, processing carcasses (i.e., meat product) with a system where the carcasses do not come into mutual contact can reduce or preclude cross-contamination. Traceability of the products throughout processing is also important. Improved heat transfer during processing is also desirable. For example, improved heat transfer during the chilling of carcasses can reduce processing time, save energy, save space, and reduce the chance of contamination. Uniform heat transfer can further improve processing by providing reliability and consistency in steps such as chilling and scalding. Additionally, having uninterrupted “inline” processing of product is valuable, because the “rehang” step of having to put conventionally immersion-chilled poultry carcasses back on to shackles has various challenges.

(3) Therefore, what is needed is system and method for inline processing that includes both rotational and translational movement of products through the processing steps.

### SUMMARY

(4) The present disclosure relates to processing systems and methods. The disclosed technology includes a system for rotating a product during processing. The system for rotating a product during processing can include a shackle system. The shackle system can include a trolley, a rotator, and a rotating shackle. The shackle system can include a track. The trolley can be coupled to the track and configured to undergo translational motion along the track causing the shackle system to move along the track. The rotator can be configured to undergo rotational kinematics causing rotation of the rotating shackle.

(5) The rotation of the rotating shackle can be caused, at least in part, by the track causing the rotational kinematics on the rotator as the shackle system moves along the track.

(6) The rotator can include a gear. The track can include a rack. The gear can mesh with the rack causing the rotational kinematics on the rotator as the shackle system moves along the track.

(7) The rotator can include a peg. The track can include a peg. The peg of the rotator can interface with the peg of the track causing the rotational kinematics on the rotator as the shackle system moves along the track.

(8) The rotation of the rotating shackle can be caused, at least in part, by a motor configured to induce the rotational kinematics on the rotator.

(9) The rotation of the rotating shackle can include a churning pattern wherein the rotator is configured to undergo alternating rotational kinematics causing the rotating shackle to rotate back and forth in a clockwise and counterclockwise direction.

(10) The shackle system can include a stabilizing rod. The stabilizing rod can be configured to extend through a cavity of the product carried by the shackle system.

- (11) The stabilizing rod can be extendable and retractable.
- (12) The shackle system can include an external entrapment. The external entrapment can be configured to extend around at least a portion of the exterior of the product.
- (13) The external entrapment can be extendable and retractable.
- (14) The shackle system can include a brake. The brake can be configured to halt the rotation of the rotating shackle.
- (15) The product can be poultry.
- (16) The disclosed technology includes a method for processing a product. The method can include attaching the product to a shackle system wherein the shackle system is coupled to a track. The method can include moving the product, with the shackle system, along the track. The method can include rotating the product, with the shackle system.
- (17) The shackle system can include a trolley, a rotator, and a rotating shackle.
- (18) The method can include causing, at least in part by the track, rotational kinematics on the rotator as the shackle system moves along the track.
- (19) The rotator can include a gear. The track can include a rack. The gear can mesh with the rack causing rotational kinematics on the rotator as the shackle system moves along the track.
- (20) The rotator can include a peg. The track can include a peg. The peg of the rotator can interface with the peg of the track causing rotational kinematics on the rotator as the shackle system moves along the track.
- (21) The method can include causing, at least in part by a motor, rotational kinematics on the rotator.
- (22) The rotation of the rotating shackle can include a churning pattern wherein the rotator is configured to undergo alternating rotational kinematics causing the rotating shackle to rotate back and forth in a clockwise and counterclockwise direction.
- (23) The method can include halting, with a brake, the rotation of the rotating shackle.
- (24) The method can include holding in place, with a brake, the orientation of the product on the shackle system.
- (25) The shackle system further can include a stabilizing rod. The method can include extending, through a cavity of the product, the stabilizing rod. The method can include retracting, through the cavity of the product, the stabilizing rod.
- (26) The shackle system can include an external entrapment. The method can include extending, around at least a portion of the exterior of the product, the external entrapment. The method can include retracting the external entrapment.
- (27) The method can include exposing at least a portion of the product to a heat transfer media, such that heat transfer occurs between the product and the heat transfer media.
- (28) The heat transfer media can be cold air.
- (29) The heat transfer media can be chilled water.
- (30) The heat transfer media can be ice slurry.
- (31) The heat transfer media can be hot water.
- (32) The processing can include primary and secondary processing of the product.
- (33) The product can be poultry.
- (34) These and other aspects of the present disclosure are described in the Detailed Description below and the accompanying drawings. Other aspects and features of embodiments will become apparent to those of ordinary skill in the art upon reviewing the following description of specific, exemplary embodiments in concert with the drawings. While features of the present disclosure may be discussed relative to certain embodiments and figures, all embodiments of the present disclosure can include one or more of the features discussed herein. Further, while one or more embodiments may be discussed as having certain advantageous features, one or more of such features can also be used with the various embodiments discussed herein. In similar fashion, while exemplary embodiments may be discussed below as device, system, or method embodiments, it is to be

understood that such exemplary embodiments can be implemented in various devices, systems, and methods of the present disclosure.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following detailed description of specific embodiments of the disclosure will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the disclosure, specific embodiments are shown in the drawings. It should be understood, however, that the disclosure is not limited to the precise arrangements and instrumentalities of the embodiments shown in the drawings.
- (2) FIG. 1 provides a rotating shackle, in accordance with the present disclosure.
- (3) FIG. 2A provides a rotating shackle, in accordance with the present disclosure.
- (4) FIG. 2B provides a rotating shackle, in accordance with the present disclosure.
- (5) FIG. 2C provides a rotating shackle, in accordance with the present disclosure.
- (6) FIG. 3 provides photos of an example system for rotating a product during processing, in accordance with the present disclosure.
- (7) FIG. 4 provides photos of an example system for rotating a product during processing, in accordance with the present disclosure.
- (8) FIG. 5 provides photos of an example system for rotating a product during processing, in accordance with the present disclosure.
- (9) FIG. 6 provides photos of an example system for rotating a product during processing, in accordance with the present disclosure.
- (10) FIG. 7 provides photos of an example system for rotating a product during processing, in accordance with the present disclosure.
- (11) FIG. 8A provides a component of a rotating shackle, in accordance with the present disclosure.
- (12) FIG. 8B provides an illustration of an example system for rotating a product during processing, in accordance with the present disclosure.
- (13) FIG. 9 provides an illustration of an example system for rotating a product during processing, in accordance with the present disclosure.
- (14) FIG. 10 provides an illustration of an example system for rotating a product during processing, in accordance with the present disclosure.

### DETAILED DESCRIPTION

- (15) Throughout this disclosure we describe systems and methods for processing, such as, a system and method for rotating a product during processing. For example, an embodiment of the disclosure provides a rotating shackle system for enhanced inline poultry processing.
- (16) While the disclosed technology is described throughout this disclosure in relation to systems and methods for processing poultry, those having skill in the art will recognize that the disclosed technology is not so limited and can be applicable to other scenarios and applications. For example, it is contemplated that the disclosed technology can be applicable to other food processing applications such as pork, beef, and other livestock.
- (17) Some implementations of the disclosed technology will be described more fully with reference to the accompanying drawings. This disclosed technology may, however, be embodied in many different forms and should not be construed as limited to the implementations set forth herein. The components described hereinafter as making up various elements of the disclosed technology are intended to be illustrative and not restrictive. Indeed, it is to be understood that other examples are contemplated. Many suitable components that would perform the same or similar functions as components described herein are intended to be embraced within the scope of the disclosed devices and methods. Such other components not described herein may include, but are not limited to, for

example, components developed after development of the disclosed technology.

(18) Herein, the use of terms such as “having,” “has,” “including,” or “includes” are open-ended and are intended to have the same meaning as terms such as “comprising” or “comprises” and not preclude the presence of other structure, material, or acts. Similarly, though the use of terms such as “can” or “may” are intended to be open-ended and to reflect that structure, material, or acts are not necessary, the failure to use such terms is not intended to reflect that structure, material, or acts are essential. To the extent that structure, material, or acts are presently considered to be essential, they are identified as such.

(19) It is to be understood that the mention of one or more method steps does not preclude the presence of additional method steps or intervening method steps between those steps expressly identified. Similarly, it is also to be understood that the mention of one or more components in a device or system does not preclude the presence of additional components or intervening components between those components expressly identified. Further, it is contemplated that the disclosed methods and processes can include, but do not necessarily include, all steps discussed herein. That is, methods and processes in accordance with the disclosed technology can include some of the disclosed while omitting others.

(20) Throughout the specification and the claims, the following terms take at least the meanings explicitly associated herein, unless otherwise indicated. The term “or” is intended to mean an inclusive “or.” Further, the terms “a,” “an,” and “the” are intended to mean one or more unless specified otherwise or clear from the context to be directed to a singular form. By “comprising,” “containing,” or “including” it is meant that at least the named element, or method step is present in article or method, but does not exclude the presence of other elements or method steps, even if the other such elements or method steps have the same function as what is named.

(21) As used herein, unless otherwise specified, the use of the ordinal adjectives “first,” “second,” “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to, and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking, or in any other manner.

(22) Although the disclosed technology may be described herein with respect to various systems and methods, it is contemplated that embodiments or implementations of the disclosed technology with identical or substantially similar features may alternatively be implemented as methods or systems. For example, any aspects, elements, features, or the like described herein with respect to a method can be equally attributable to a system. As another example, any aspects, elements, features, or the like described herein with respect to a system can be equally attributable to a method.

(23) Reference will now be made in detail to examples of the disclosed technology, examples of which are illustrated in the accompanying drawings and disclosed herein. Wherever convenient, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

(24) Referring now to the drawings, in which like numerals represent like elements, examples of the present disclosure are herein described. As will be described in greater detail, the present disclosure can include a system and method for rotating products during processing. To provide a background of the system described in the present disclosure, components of the system for rotating products during processing is shown in FIG. 1 and will be discussed first.

(25) To facilitate an understanding of the principles and features of the present disclosure, various examples of the disclosed technology are explained herein. The components, steps, and materials described herein as making up various elements of the disclosed technology are intended to be illustrative and not restrictive. Many suitable components, steps, and materials that would perform the same or similar functions as the components, steps, and materials described herein are intended to be embraced within the scope of the disclosure. Such other components, steps, and materials not described herein can include, but are not limited to, similar components or steps that are developed after development of the embodiments disclosed herein.

(26) As shown in FIG. 1, the disclosed technology includes a shackle system **100** for rotating products during processing. The shackle system **100** can include a body **102**. For example, the body **102** can be a slender rod or tube. The shackle system **100** can include a trolley **110**. For example, the trolley **110** can be disposed at a proximal end of the body **102** and can be configured to engage or couple with the track of a processing line. Alternatively, or in addition, the shackle system **100** can include a rotator **120** configured to rotate a product attached to the rotating shackle. For example, the rotator **120** can include a gear **122** and a rotatable cylinder **124**. Alternatively, or in addition, the shackle system **100** can include a rotating shackle **130**. For example, the rotating shackle **130** can be configured to receive and hold a product for processing (e.g., a poultry carcass). Alternatively, or in addition, the shackle system **100** can include a stabilizing rod **140**. For example, the stabilizing rod **140** can be disposed at a distal end of the body **102** and can extend through the product to keep the product center of mass aligned with the vertical axis of the shackle assembly.

(27) The trolley **110** can include one or more wheels configured to move the shackle system **100** along a track of a processing line. For example, the trolley **110** can include a first wheel **112** and a second wheel **114** configured to engage the track of a processing line wherein the wheels are disposed on the top of the track allowing the rotating shackle to hang from the track. Alternatively, or in addition, the trolley **110** can include any movement structure known in the art, including, but not limited to wheels, bearings, sliders, gears, tracks, and the like, or any combination thereof. The trolley **110** can be securely attached to the proximate end of the body **102**. Alternatively, or in addition, the trolley **110** can be moveably attached to the proximate end of the body **102**. For example, the trolley **110** can attach to the body **102** with a hinge to allow the body **102** to freely swing. Alternatively, the trolley **110** can be removably attached to the proximate end of the body **102**.

(28) The rotator **120** can be configured to rotate the rotating shackle **130** as the shackle system **100** moves along a processing track. For example, the rotator **120** can undergo rotational kinematics when the shackle system **100** undergoes translational motion along the track. The rotator can include a gear **122** configured to mesh with teeth along a processing track wherein the teeth of the track induce rotation on the gear **122** as the shackle system **100** moves along the track. Alternatively, or in addition, the rotator **120** can include any rotator known in the art, including, but not limited to a gear, sprocket, pulley, pinion, wheel, peg, and the like, or any combination thereof, wherein a structure of the track induces rotation on the rotator **120** as the shackle system **100** moves along the track. Alternatively, or in addition, the rotator **120** can include an actuator configured to actively rotate the rotating shackle **130**. For example, the rotator can include a motor configured to controllably rotate, hold, and position the rotating shackle **130**.

(29) The rotatable cylinder **124** can be a hollow cylinder with an interior configured to fit the body **102** of the shackle system **100**. In addition, the rotatable cylinder **124** can include bearings on the interior to allow for free rotation around the body **102** passing through.

(30) The rotating shackle **130** can be any shackle known in the art, including, but not limited to a shackle hook, slip hook, evisceration shackle, meat hook, poultry shackle, pork shackle, beef shackle, or any other type of livestock shackle or hook, and the like, or any combination thereof. The rotating shackle **130** can be attached to rotator **120** with a fastener **132**. For example, as illustrated in FIG. 1, the rotating shackle **130** can be attached to the rotatable cylinder **124** with a mounting plate and U-bolt for a secure attachment.

(31) The stabilizing rod **140** can be any elongated member that provides additional mass in line with the vertical axis of the rotating shackle. For example, a metal rod. In addition, the stabilizing rod **140** can be extendable and retractable. For example, the stabilizing rod **140** can be retracted to allow for a product to be placed on or removed from rotating shackle **130** and can be extended, through the cavity of the product, once the product is in place on the rotating shackle **130**. Alternatively, or in addition, the stabilizing rod can be an extension of the body **102**. For example, the body **102** can extend from the trolley **110** attached to the proximal end of the body **102**, through

the rotator **120**, and down through the product.

(32) As shown in FIGS. 2A-C, the rotating shackle can include an external structure. For example, when a product does not have a cavity, such as a cavity of a poultry carcass, the rotating shackle can include an external structure, alternatively, or in addition to, a stabilizing rod. The external structure can be an external entrapment **210**. For example, the external entrapment **210** can include one or more structures disposed at a distal end of the body **102** configured to be positioned around the exterior of the product to help keep the product center of mass aligned with the vertical axis of the shackle assembly. The external entrapment **210** can comprise a plurality of rods extending down around the exterior of the product. For example, the plurality of rods can generally form a cage around the product holding the product in place. Alternatively, or in addition, the rods can be flexible to accommodate products with varying sizes. In addition, the external entrapment **210** can be extendable and retractable.

(33) As illustrated in FIG. 2B, the external entrapment can be extended to an engaged position **230**. For example, the external entrapment **210** can be extended down around the exterior of the product. In addition, as illustrated in FIG. 2C, the external entrapment can be retracted to a retracted position **240**. For example, the external entrapment **210** can be retracted up away from the product to allow the product to be placed onto or removed from the rotating shackle **130**.

(34) Alternatively, or in addition, the shackle system **100** can include a brake **220**. The brake **220** can be configured to stop or prevent rotation of the rotating shackle. Additionally, the brake can be configured to hold the rotating shackle in one or more distinct positions or orientations. For example, the brake can hold the rotating shackle in a position for cutting the product.

(35) As shown in FIG. 3, the disclosed technology can include a processing system **300**. The processing system **300** can include a track **310**. For example, the track **310** can include an overhead track wherein one or more shackle systems **100** hang from the overhead track and can move along the overhead track. The track **310** can be a continuous track that spans the entire processing process. For example, the track **310** can be continuous through the primary and secondary processing (e.g., slicing, separating, cutting) of the product. Alternatively, or in addition, the track **310** can span a portion of the processing process.

(36) As shown in FIG. 4, the processing system **300** can include a plurality of shackle systems **100**. The shackle system **100** can hold a product **410**. For example, as illustrated in FIG. 4, the product **410** can be a poultry carcass.

(37) As shown in FIG. 4, the shackle system **100** can be coupled to the track **310** by the trolley **110**. The trolley system **110** can roll along the track causing translation motion of the shackle system **100**.

(38) Alternatively, or in addition, as shown in FIG. 4, the processing system **300** can include one or more links **420** between shackle systems **100**. The links **420** can assist the product **410** in keeping near-vertical orientation. For example, the links **420** can provide force to hold the product in place vertically and counteract any drag effect caused by the chilling medium. Alternatively, or in addition, the links **420** can translate rotational motion from a shackle system **100** to an adjacent shackle system **100**. For example, the links **420** can translate rotational motion from a shackle system **100** with a rotator **120** to an adjacent shackle system **100**.

(39) As shown in FIG. 5, the processing system **300** can include one or more secondary tracks **510**. For example, the secondary track **510** can be a side track disposed at a height such that it interacts with the rotator **120**. Alternatively, or in addition, the secondary track **510** can be disposed in line with the track **310** or can run parallel with the track **310**. The secondary track **510** can include one or more structures that interact with the rotator **120** and cause rotation of the rotator **120** as the shackle system **100** moves along the track. For example, as illustrated in FIG. 5, the secondary track **510** can include a chain **520** and the rotator **120** can include a sprocket **530**. As the shackle system **100**, via the trolley **110**, moves along the track **310**, the chain **520** can mesh with the sprocket **530** causing rotation of the rotator **120** and rotating shackle **130**.



(40) As shown in FIG. 6, the secondary track **510** can include a first side track **612** and a second side track **614**. The first and second side tracks **612** and **614** can be disposed on either side of the shackle system **100**. The first and second side tracks **612** and **614** can be disposed at a height such that they interact with the rotator **120**. The first and second side tracks **612** and **614** can include one or more structures that interact with the rotator **120** and cause rotation of the rotator **120** as the shackle system **100** moves along the track. For example, as illustrated in FIG. 6, the second side track **614** can include a rack **620** and the rotator **120** can include a gear **122**. As the shackle system **100**, via the trolley **110**, moves along the track, the rack **620** can mesh with the gear **122** causing rotation of the rotator **120**. The rack **620** can include a plurality of pins **622**. For example, the pins **622** can be sized and spaced to allow for the gear **122**, to mesh. Alternatively, or in addition, the first side track **612** can include a rack.

(41) As shown in FIG. 7, the first side track **612** and the second side track **614** can include alternating sections of rack **620**. The sections of rack **620** can comprise a plurality of pins **622**. In addition, the first side track **612** and the second side track **614** can include alternating sections of blank **710**. For example, blank **710** can be a solid material block such as ultra-high molecular-weight polyethylene. The blank **710** can be located in areas where there are no pins **622** to allow for proper gear meshing on the opposite side where there are pins **622**. For example, as illustrated in FIG. 7, the first side track **612** and second side track **614** can include alternating sections of rack **620** and blank **710** wherein the sections of rack **620** of the first side track **612** align with the sections of blank **710** of the second side track **614** and vice versa.

(42) Alternatively, or in addition, as shown in FIG. 8A, the rotator **120** can include one or more pegs **810**. Alternatively, or in addition, the secondary track **510** include one or more track pegs **820**. As illustrated in FIG. 8B, the track pegs **820** can interact (e.g., interface, contact) with the pegs **810** of the rotator **120** and cause rotation of the rotator **120** as the shackle system **100** moves along the track.

(43) Alternatively, or in addition, the secondary track **510** and the rotator **120** can include any structures or systems known in the art to cause rotational motion, including but not limited to, gears, wheels, rack and pinion, pegs, chain and sprocket, and the like, or any combination thereof. Alternatively, or in addition, the rotating shackle **130** can be rotated by any movement method known in the art, including but not limited to, direct drive, pneumatic, hydraulic, serial manipulator, cable drive, gears, wheels, pulleys, rack and pinion, pegs, chain and sprocket, and the like, or any combination thereof. The rotation of the rotating shackle **130** can be independent of the translational movement of the shackle system **100** along the track **310**.

(44) As shown in FIGS. 9 and 10, the processing system **300** can include moving and rotating a plurality of products **410** through a heat transfer media **910**. The rotational pattern of the products **410** through the heat transfer media **910** can allow better control and enhanced transport phenomena between the product **410** and heat transfer media **910**. The heat transfer media **910** can be any medium with a temperature differential to the product **410**. The heat transfer media **910** can include a chilled media for chilling the products **410**. For example, the heat transfer media **910** can include chilled air, chilled water, ice slurry, and the like, or any combination thereof. Alternatively, or in addition, the heat transfer media **910** can include a heated media for heating the products **410**. For example, hot water for scalding the products **410**.

(45) The rotational pattern can be a churning pattern, wherein the product **410** counter-rotates before completing a revolution. Alternately, or in addition, the rotational pattern can include consistent single direction rotation (e.g., clockwise or counterclockwise), as illustrated in FIGS. 9 and 10. Alternately, or in addition, the rotational pattern can include single direction rotation with intermittent stops in rotation.

(46) The processing system **300** can include a pattern wherein the product **410** passes through the heat transfer media **910** one or more times. For example, as illustrated in FIGS. 9 and 10, the products **410** can pass through the heat transfer media **910** in a first direction and then back through

the heat transfer media **910** in a second direction. The first direction can be generally opposite to the second direction.

(47) The following examples further illustrate aspects of the present disclosure. However, they are in no way a limitation of the teachings or disclosure of the present disclosure as set forth herein.

#### EXAMPLES

(48) In one embodiment, the invention includes a poultry carcass conveyor system that includes rotating carcass shackles that cause poultry carcasses to rotate about a vertical axis while the carcasses are in a chiller medium, such as water. The rotating shackles are attached to a conveyor that passes the carcasses through the chiller medium. Rotation of the carcasses results in quicker and more uniform internal cooling of each carcass.

(49) Optimizing the rotational carcass kinematic (i.e., that is superimposed upon the default translational kinematic) is an important sensitivity. As a leading example, “churning” patterns, wherein carcasses counter-rotate before completing a revolution, have generally been found to be more beneficial to chilling than rotational patterns wherein carcasses rotate multiple revolutions before changing direction. Various factors contribute to this added enhancement via churning, such as more intense boundary layer disruption.

(50) Optimally rotating shackles are projected to allow better control and enhanced transport phenomena between the poultry carcasses and chiller media (e.g., chilled water, ice slurry). Rotating shackles can further allow increased mass transfer (e.g., water intake into the product) during processing. Inline approaches already have key food safety features such as (1) reduced/precluded cross-contamination via carcasses not coming into mutual contact and greater traceability opportunities; (2) Rotating shackles provide an additional and controlled contribution to relative velocity between the carcasses and chiller medium. This then means that key convective heat transfer parameters such as Nusselt number are not limited by line speed (e.g., 140 birds per minute), because rotational vectors offer another kinematic degree-of-freedom; (3) Optimized rotational patterns by preceding carcasses may reduce wake effects seen by trailing carcasses, hence allowing for better uniformity of chill of the sequence of carcasses; (4) Specific to the ice slurry, issues of variations in carcasses exposure to ice (e.g., due to ice stratification and variable path lines of unconstrained carcasses) is addressed via consistent carcass depths and pathways within the chiller medium; (5) Specific to the ice slurry, rotating carcasses may serve as “agitators” to help mechanically reduce agglomeration as opposed to having complete dependence upon a freezing point depressant/dispersing agent such as a salt.

(51) Another example includes a practical shackle carousel system with induced active and passive-mechanical shackle rotation. Passive shackle rotational systems include shackle rotation that is coupled with the processing line speed. The applications of this rotational shackle system can be integrated throughout the poultry processing plant including first and secondary processing to improve productivity. Furthermore, at the end of rotation, the rotating shackle can be held in desired orientation by brake or locking the system. Active methods of inducing shackle rotation that are decoupled from the processing line speed that rotate independent of processing line translation (i.e. motor-driven pulley, electronic shackle control) can be implemented in a similar manner as shown.

(52) Rotating Shackle System Proof-of-Concept Development

(53) Significant time and effort was put in to building a practical shackle carousel system with induced active and passive-mechanical shackle rotation. Active methods of inducing shackle rotation that are decoupled from the processing line speed that rotate independent of processing line translation (i.e. motor-driven pulley, electronic shackle control) can be implemented in a similar manner as disclosed. Passive shackle rotational systems include shackle rotation that is coupled with the processing line speed. The applications of this rotational shackle system can be integrated throughout the poultry processing plant including first and secondary processing to improve productivity. Furthermore, at the end of rotation, the rotating shackle can be held in desired

orientation by brake or locking the system.

#### (54) Carousel Test Rig Subsystem

(55) FIG. 3 displays the entire shackle carousel system and FIG. 4 displays a portion of the carousel shackle line. The carousel track is 280.1 inches (~23.3 ft) in total length with two straight track lengths of 111 inches and two half-circle bends with diameters of 18.5 inches each. The test rig capacity was modified to hold a maximum capacity of 593.8 gallons chiller media with a width of 36 inches, length of 161 inches, and a height of 24 inches. The test rig was modified in this manner to accommodate the standard, industrial 18.5-inch diameter pulleys and allow for the shackle line to be moving within the center of the channel at all times. The shackle carousel line is driven by a 1 HP SEW EuroDrive motor and motor VFD controller used in conjunction with a timing belt that is coupled to the driver pulley. The driver pulley has notches where the shackle trolleys (linked by Grade 30 chain and spaced 12 inches apart) fall into place and are transported around the carousel track.

(56) With the motor VFD control, the carousel line speed can be varied up to 3.3 ft/s. Based upon typical industrial processing speeds of 130-140 birds per minute (which translates to a high end speed of 2.2-2.3 ft./s), there is more than enough motor power to drive the system.

#### (57) Modified Rotational Shackle Subsystem

(58) In lieu of the motorized shackle that is programmed to rotate in a certain manner or similar translation-independent rotation method, a mechanical rotational shackle assembly, displayed in FIG. 1, was designed and constructed as one proof-of-concept method for implementing passively-induced shackle rotation. The modified assembly implements additional rotational shackle kinematics to the traditional translational shackle motion by way of the pinion gear. In this assembly, the pinion gear can be varied in size depending on the desired shackle RPM for a fixed line speed. The diameter of the pinion gear in the current proof-of-concept is 6.4 inches, which corresponds to shackle RPMs of 70-75 RPM based upon industrial translation line speeds of 2.0-2.2 ft./s. As mentioned, the relationship between shackle angular velocity and translational velocity is fixed due to the pinion gear sizing, but it can be varied with replacement by a smaller/larger gear. Another feature of the rotational shackle assembly is the method of holding the chicken in place with a stabilizing rod that extends beyond the shackle when the chicken has a middle cavity. This helps the system to keep the carcass center of mass in line with the shackle center axis as it rotates back and forth. Without this stabilization, the carcass's rotational inertia may displace it from the central shackle axis and can cause the carcass to fall off the shackle. A chicken with a cavity can be held in place with a stabilizing center rod that extends beyond the shackle. For other processing steps, in which the chicken does not have a cavity, the chicken can be held stable with an exterior entrapment, and one proof-of-concept design can be seen in FIGS. 2A-C. In certain steps of processing, shackle orientation is critical to process efficacy so methods for maintaining proper shackle orientation have been devised. Shackle rotation brake, lock, or pin extension methods to trigger a stop can be implemented, and one proof-of-concept design can be seen in FIGS. 2A-C.

#### (59) Rotational Guidance Track System

(60) To complete the passive-mechanical rotating shackle system, a guidance track system consisting of four 7-foot straight aluminum u-channels (two for each flow channel) with 127 holes each are arranged in the plane of the rotating pinion gear to serve as a track for the prescribed rotation. The drilled holes along the length of the u-channel allowed for a robust, custom track to be configured with the bolts and pins as can be seen in FIGS. 6 and 7. In another example of the track construction, a standard bike chain, as can be seen in FIG. 5, was used to pair with the pinion gear. To configure for better track guidance, a robust u-channel track has a greater channel height to accommodate slight pinion tilts, and this method allows for the rotational method and length of rotation to be varied as the operator desires. If churning patterns are desired, then the pins within the u-channels can be configured to have alternating segments of pins and no pins. With the same rationale, continuous rotation can be achieved by arranging one u-channel side with all pins while

having no pins on the other u-channel side. As can be seen in FIG. 7, the u-channel is lined with blue strips of ultra-high molecular-weight polyethylene for the areas where there are no pins to allow for proper gear meshing on the opposite side where there are pins. Without the low-friction padding as a backing, the pinion gear will attempt to “push out” when it encounters a sequence of pins.

(61) It is to be understood that the embodiments and claims disclosed herein are not limited in their application to the details of construction and arrangement of the components set forth in the description and illustrated in the drawings. Rather, the description and the drawings provide examples of the embodiments envisioned. The embodiments and claims disclosed herein are further capable of other embodiments and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purposes of description and should not be regarded as limiting the claims.

(62) Accordingly, those skilled in the art will appreciate that the conception upon which the application and claims are based may be readily utilized as a basis for the design of other structures, methods, and systems for carrying out the several purposes of the embodiments and claims presented in this application. It is important, therefore, that the claims be regarded as including such equivalent constructions.

(63) Furthermore, the purpose of the Abstract is to enable the United States Patent and Trademark Office and the public generally, and especially including the practitioners in the art who are not familiar with patent and legal terms or phraseology, to determine quickly from a cursory inspection the nature and essence of the technical disclosure of the application. The Abstract is neither intended to define the claims of the application, nor is it intended to be limiting to the scope of the claims in any way.

## Claims

1. A system comprising: a shackle system comprising: a rotator; and a rotating shackle; and a track; wherein: the rotator is configured to undergo rotational kinematics causing rotation of the rotating shackle; and the rotation of the rotating shackle is caused, at least in part, by a motor configured to induce the rotational kinematics on the rotator.
2. The system of claim 1, wherein the rotation of the rotating shackle is further caused, at least in part, by the track causing the rotational kinematics on the rotator as the shackle system moves along the track.
3. The system of claim 2, wherein: the rotator comprises a gear; the track comprises a rack; and the gear meshes with the rack causing the rotational kinematics on the rotator as the shackle system moves along the track.
4. The system of claim 1, wherein: the rotator comprises a rotator peg; the track comprises a track peg; and the rotator peg interfaces with the track peg causing the rotational kinematics on the rotator as the shackle system moves along the track.
5. The system of claim 1, wherein the rotator is further configured to undergo alternating rotational kinematics causing rotation of the rotating shackle back and forth in a clockwise and counterclockwise direction.
6. The system of claim 1 further comprising a stabilizing rod configured to extend through a cavity of a product carried by the shackle system; wherein the stabilizing rod is extendable and retractable.
7. The system of claim 1 further comprising an external entrapment configured to extend around at least a portion of an exterior of a product carried by the shackle system; wherein the external entrapment is extendable and retractable.
8. The system of claim 1, wherein the shackle system further comprises a trolley; wherein the trolley is coupled to the track and configured to undergo translational motion along the track

causing the shackle system to move along the track; and wherein at least one of: (i) the rotator comprises a rotator peg, the track comprises a track peg, and the rotator peg interfaces with the track peg causing the rotational kinematics on the rotator as the shackle system moves along the track; (ii) the system further comprises a stabilizing rod configured to extend through a cavity of a product carried by the shackle system, wherein the stabilizing rod is extendable and retractable; or (iii) the system further comprises an external entrapment configured to extend around at least a portion of an exterior of a product carried by the shackle system, wherein the external entrapment is extendable and retractable.

9. The system of claim 8, wherein the product is poultry.

10. A method for processing a product with the system of claim 1 comprising: attaching the product to the shackle system coupled to the track; moving the product, with the shackle system, along the track; and rotating the product at least in part by the motor.

11. The method of claim 10 further comprising: rotating the product at least in part by the track causing the rotational kinematics on the rotator as the shackle system moves along the track; and transferring thermal energy between at least a portion of the product and a heat transfer media.

12. The method of claim 11, wherein: the rotator comprises a gear; the track comprises a rack; and rotating the product, at least in part by the track, comprises meshing the gear with the rack.

13. The method of claim 11, wherein: the rotator comprises a rotator peg; the track comprises a track peg; and rotating the product, at least in part by the track, comprises interfacing the rotator peg with the rack peg.

14. The method of claim 11, wherein the heat transfer media is selected from the group consisting of cold air, chilled water, ice slurry, hot water, and a combination thereof.

15. The method of claim 10, wherein the rotator is further configured to undergo alternating rotational kinematics causing rotation of the rotating shackle back and forth in a clockwise and counterclockwise direction.

16. The method of claim 10 further comprising: halting, with a brake, the rotation of the rotating shackle.

17. The method of claim 10 further comprising: holding in place, with a brake, an orientation of the product on the shackle system.

18. The method of claim 10 further comprising: extending, through a cavity of the product, a stabilizing rod of the shackle system; and retracting, through the cavity of the product, the stabilizing rod.

19. The method of claim 10 further comprising: extending, around at least a portion of an exterior of the product, an external entrapment of the shackle system; and retracting the external entrapment.

20. The system of claim 1, wherein the system is configured for rotating a product during processing.

21. The system of claim 1 further comprising a brake configured to halt the rotation of the rotating shackle.

22. The system of claim 20, wherein the product is poultry.

23. The system of claim 20 further comprising a heat transfer media; wherein the system is further configured to transfer thermal energy between at least a portion of the product and the heat transfer media.

24. The system of claim 23, wherein the heat transfer media is selected from the group consisting of cold air, chilled water, ice slurry, hot water, and a combination thereof.

25. A system comprising: a shackle system comprising: a rotator comprising a rotator peg; and a rotating shackle; and a track comprising a track peg; wherein the rotator peg interfaces with the track peg causing the rotator to undergo rotational kinematics as the shackle system moves along the track causing rotation of the rotating shackle.

26. A method for processing a product with a system comprising a shackle system coupled to a

track, the shackle system comprising a rotator, a rotating shackle, and a stabilizing rod, wherein the rotator is configured to undergo rotational kinematics causing rotation of the rotating shackle, the method comprising: extending, through a cavity of the product, the stabilizing rod of the shackle system to attach the product to the shackle system; moving the product, with the shackle system, along the track; rotating the product with the shackle system; and retracting, through the cavity of the product, the stabilizing rod.

27. A method for processing a product with a system comprising a shackle system coupled to a track, the shackle system comprising a rotator, a rotating shackle, and an external entrapment, wherein the rotator is configured to undergo rotational kinematics causing rotation of the rotating shackle, the method comprising: extending, around at least a portion of an exterior of the product, the external entrapment of the shackle system; moving the product, with the shackle system, along the track; rotating the product with the shackle system; and retracting the external entrapment.

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